```
clc;
clear;
close all;

%% ============================================================
% EXPERIMENT:
% DESIGN AND ANALYSIS OF FIR DIGITAL FILTERS
% USING FOURIER SERIES AND WINDOWING TECHNIQUES
%
% CO Mapping : CO2
% Bloom's Level : K4 - Analyze
% WK Level : WK3 - Engineering Fundamentals
%% ============================================================

%% 1. DESIGN SPECIFICATIONS

Fs = 1000;          % Sampling frequency (Hz)
Fc = 150;           % Cutoff frequency (Hz)
N  = 51;            % Number of FIR filter coefficients
                    % Choose an odd number for Type-I FIR

M = (N - 1)/2;      % Filter midpoint
n = 0:N-1;          % Sample index

Wc = 2*pi*Fc/Fs;    % Normalized cutoff frequency (rad/sample)

fprintf('\n===========================================\n');
```

```
===========================================
```

```
fprintf(' FIR DIGITAL FILTER DESIGN USING MATLAB\n');
```

```
 FIR DIGITAL FILTER DESIGN USING MATLAB
```

```
fprintf('===========================================\n');
```

```
===========================================
```

```
fprintf('Sampling Frequency = %d Hz\n', Fs);
```

```
Sampling Frequency = 1000 Hz
```

```
fprintf('Cutoff Frequency   = %d Hz\n', Fc);
```

```
Cutoff Frequency   = 150 Hz
```

```
fprintf('Filter Length      = %d\n', N);
```

```
Filter Length      = 51
```

```
fprintf('Filter Order       = %d\n', N-1);
```

```
Filter Order       = 50
```

```
%% 2. IDEAL LOW-PASS FIR FILTER USING FOURIER SERIES

% Ideal low-pass impulse response
hd = zeros(1, N);

for i = 1:N
    k = n(i) - M;

    if k == 0
        hd(i) = Wc/pi;
    else
        hd(i) = sin(Wc*k)/(pi*k);
    end
end

% Rectangular window: equivalent to direct truncation
w_rect = rectwin(N).';

% FIR coefficients using Fourier Series truncation
b_rect = hd .* w_rect;

%% 3. DESIGN FIR FILTERS USING WINDOWING TECHNIQUES

% Hamming window
w_hamming = hamming(N).';
b_hamming = hd .* w_hamming;

% Hann (Hanning) window
w_hann = hann(N).';
b_hann = hd .* w_hann;

% Blackman window
w_blackman = blackman(N).';
b_blackman = hd .* w_blackman;

% Normalize each filter to unity gain at DC
b_rect = b_rect / sum(b_rect);
b_hamming = b_hamming / sum(b_hamming);
b_hann = b_hann / sum(b_hann);
b_blackman = b_blackman / sum(b_blackman);

%% 4. DISPLAY FILTER COEFFICIENTS

fprintf('\n==========================================\n');
```

```
==========================================
```

```
fprintf('       FIR FILTER COEFFICIENTS\n');
```

```
       FIR FILTER COEFFICIENTS
```

```
fprintf('=============================================\n');
```

```
 ============================================
```

```
disp('Ideal impulse response hd(n):');
```

```
 Ideal impulse response hd(n):
```

```
disp(hd);
```

```
    -0.0127   -0.0078    0.0043    0.0138    0.0123   -0.0000   -0.0136   -0.0168   -0.0058    0.0117    0.
```

```
disp('Rectangular window coefficients:');
```

```
 Rectangular window coefficients:
```

```
disp(w_rect);
```

```
     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1
```

```
disp('Hamming window coefficients:');
```

```
 Hamming window coefficients:
```

```
disp(w_hamming);
```

```
    0.0800    0.0836    0.0945    0.1123    0.1369    0.1679    0.2047    0.2468    0.2935    0.3441    0.
```

```
disp('Hann window coefficients:');
```

```
 Hann window coefficients:
```

```
disp(w_hann);
```

```
         0    0.0039    0.0157    0.0351    0.0618    0.0955    0.1355    0.1813    0.2321    0.2871    0.
```

```
disp('Blackman window coefficients:');
```

```
 Blackman window coefficients:
```

```
disp(w_blackman);
```

```
         0    0.0014    0.0058    0.0134    0.0247    0.0402    0.0605    0.0863    0.1180    0.1561    0.
```

```
disp('FIR coefficients - Rectangular:');
```

```
 FIR coefficients - Rectangular:
```

```
disp(b_rect);
```

```
   -0.0129   -0.0079    0.0043    0.0139    0.0124   -0.0000   -0.0137   -0.0170   -0.0059    0.0118    0.
```

```
disp('FIR coefficients - Hamming:');
```

```
FIR coefficients - Hamming:
```

```
disp(b_hamming);
```

```
   -0.0010   -0.0007    0.0004    0.0015    0.0017   -0.0000   -0.0028   -0.0042   -0.0017    0.0040    0.
```

```
disp('FIR coefficients - Hann:');
```

```
FIR coefficients - Hann:
```

```
disp(b_hann);
```

```
         0   -0.0000    0.0001    0.0005    0.0008   -0.0000   -0.0018   -0.0030   -0.0013    0.0034    0.
```

```
disp('FIR coefficients - Blackman:');
```

```
FIR coefficients - Blackman:
```

```
disp(b_blackman);
```

```
         0   -0.0000    0.0000    0.0002    0.0003   -0.0000   -0.0008   -0.0015   -0.0007    0.0018    0.
```

```
%% 5. FREQUENCY RESPONSE ANALYSIS

Nfft = 4096;

[H_rect, f] = freqz(b_rect, 1, Nfft, Fs);
[H_hamming, ~] = freqz(b_hamming, 1, Nfft, Fs);
[H_hann, ~] = freqz(b_hann, 1, Nfft, Fs);
[H_blackman, ~] = freqz(b_blackman, 1, Nfft, Fs);

% Magnitude responses in dB
Mag_rect = 20*log10(abs(H_rect) + eps);
Mag_hamming = 20*log10(abs(H_hamming) + eps);
Mag_hann = 20*log10(abs(H_hann) + eps);
Mag_blackman = 20*log10(abs(H_blackman) + eps);

%% 6. PLOT WINDOW FUNCTIONS

figure;

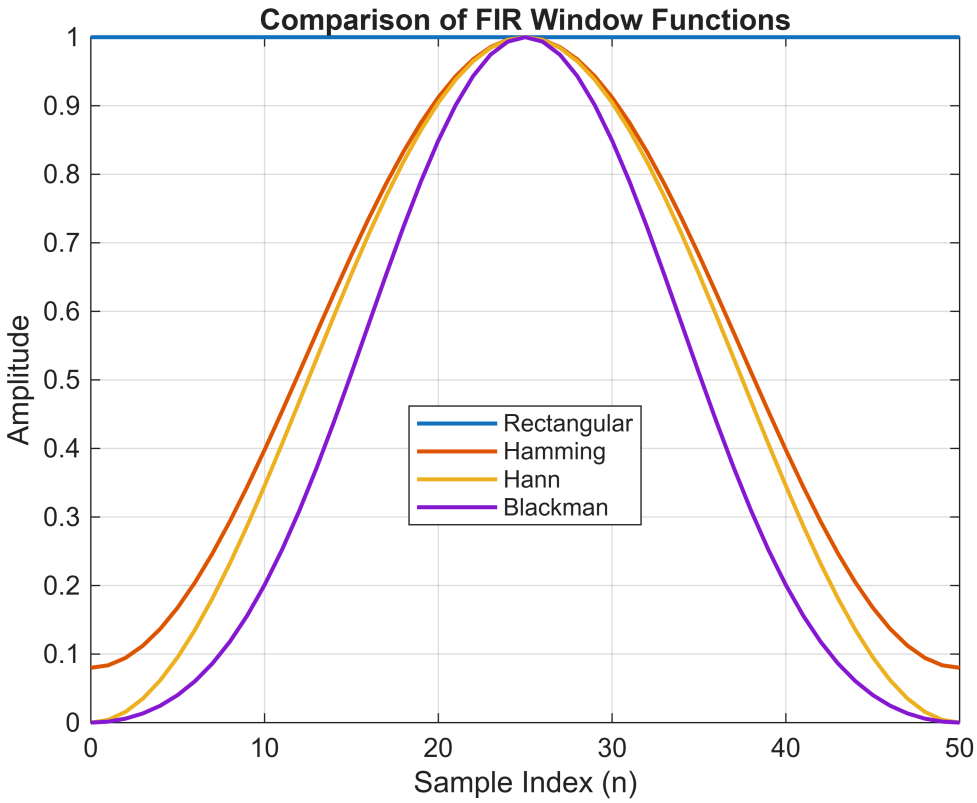
plot(n, w_rect, 'LineWidth', 1.5);
hold on;
plot(n, w_hamming, 'LineWidth', 1.5);
```

```
plot(n, w_hann, 'LineWidth', 1.5);
plot(n, w_blackman, 'LineWidth', 1.5);

grid on;
xlabel('Sample Index (n)');
ylabel('Amplitude');
title('Comparison of FIR Window Functions');

legend('Rectangular', 'Hamming', 'Hann', 'Blackman', ...
       'Location', 'best');

hold off;
```




```
%% 7. PLOT IDEAL AND WINDOWED IMPULSE RESPONSES

figure;

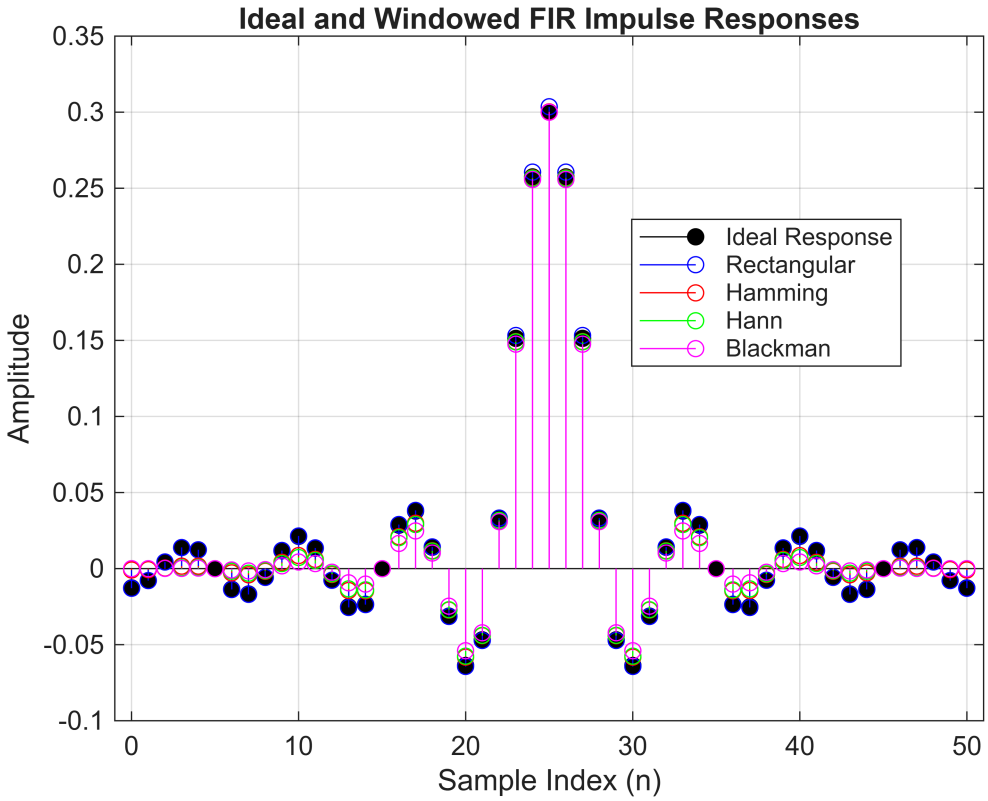
stem(n, hd, 'k', 'filled');
hold on;
stem(n, b_rect, 'b');
stem(n, b_hamming, 'r');
stem(n, b_hann, 'g');
stem(n, b_blackman, 'm');

grid on;
xlabel('Sample Index (n)');
```

```
ylabel('Amplitude');
title('Ideal and Windowed FIR Impulse Responses');

legend('Ideal Response', 'Rectangular', 'Hamming', ...
       'Hann', 'Blackman', 'Location', 'best');

hold off;
```




```
%% 8. MAGNITUDE RESPONSE COMPARISON

figure;

plot(f, Mag_rect, 'LineWidth', 1.2);
hold on;
plot(f, Mag_hamming, 'LineWidth', 1.2);
plot(f, Mag_hann, 'LineWidth', 1.2);
plot(f, Mag_blackman, 'LineWidth', 1.2);

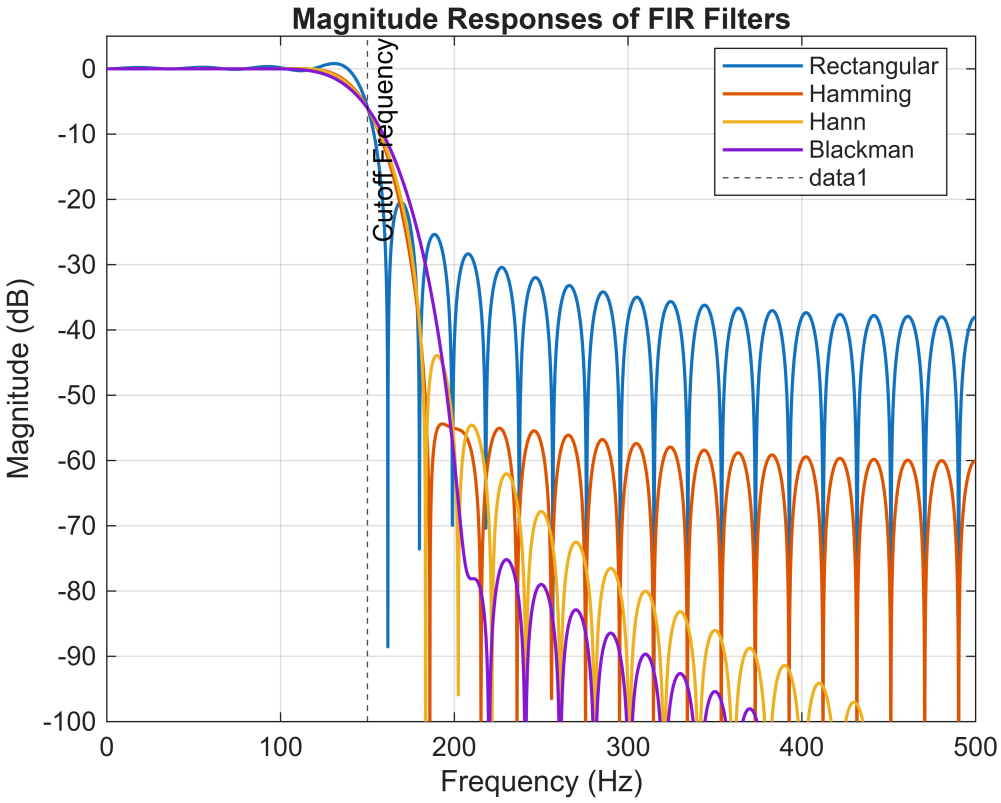
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Responses of FIR Filters');

legend('Rectangular', 'Hamming', 'Hann', 'Blackman', ...
       'Location', 'best');
```

```
xlim([0 Fs/2]);
ylim([-100 5]);

xline(Fc, '--k', 'Cutoff Frequency');

hold off;
```




```
%% 9. LINEAR MAGNITUDE RESPONSE

figure;

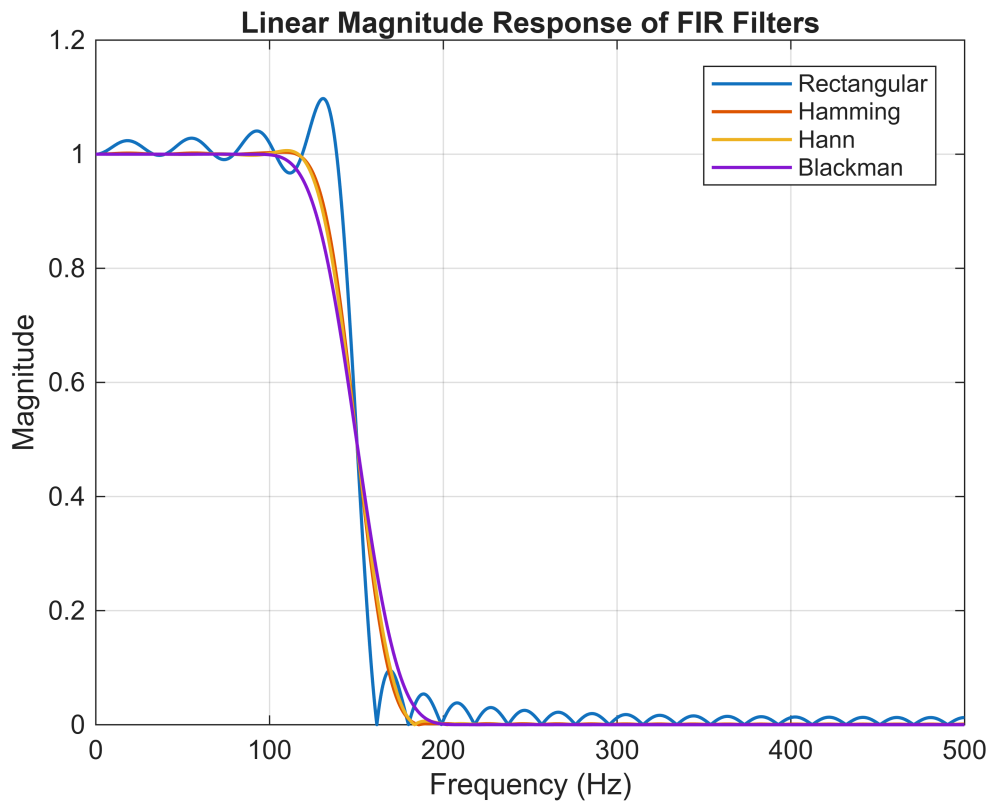
plot(f, abs(H_rect), 'LineWidth', 1.2);
hold on;
plot(f, abs(H_hamming), 'LineWidth', 1.2);
plot(f, abs(H_hann), 'LineWidth', 1.2);
plot(f, abs(H_blackman), 'LineWidth', 1.2);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Linear Magnitude Response of FIR Filters');

legend('Rectangular', 'Hamming', 'Hann', 'Blackman', ...
       'Location', 'best');

xlim([0 Fs/2]);
```

```
hold off;
```




```
%% 10. PHASE RESPONSE COMPARISON

figure;

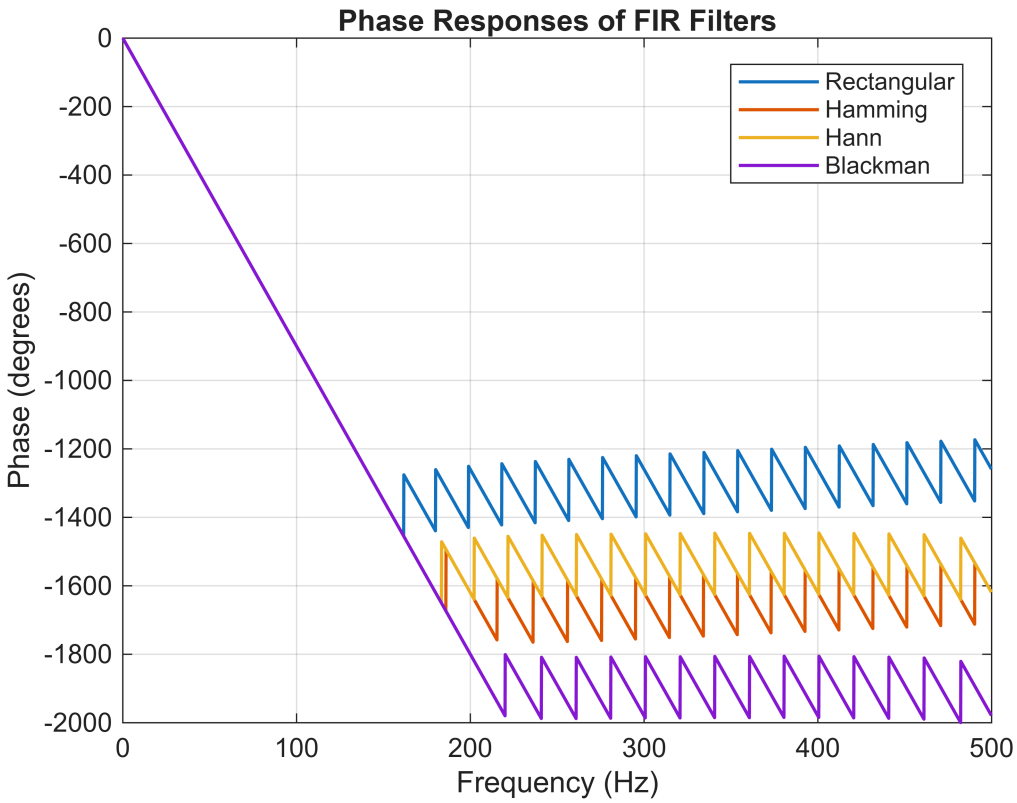
plot(f, unwrap(angle(H_rect))*180/pi, 'LineWidth', 1.2);
hold on;
plot(f, unwrap(angle(H_hamming))*180/pi, 'LineWidth', 1.2);
plot(f, unwrap(angle(H_hann))*180/pi, 'LineWidth', 1.2);
plot(f, unwrap(angle(H_blackman))*180/pi, 'LineWidth', 1.2);

grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('Phase Responses of FIR Filters');

legend('Rectangular', 'Hamming', 'Hann', 'Blackman', ...
       'Location', 'best');

xlim([0 Fs/2]);

hold off;
```




```
%% 11. FILTER PERFORMANCE COMPARISON

% Passband and stopband regions
Passband = f <= 0.8*Fc;
Stopband = f >= 1.2*Fc;

% Store magnitude responses
Mag = [Mag_rect, Mag_hamming, Mag_hann, Mag_blackman];

WindowNames = ["Rectangular";
               "Hamming";
               "Hann";
               "Blackman"];

PassbandRipple = zeros(4,1);
StopbandAttenuation = zeros(4,1);

for i = 1:4
    Hmag = Mag(:,i);

    % Peak-to-peak passband ripple in dB
    PassbandRipple(i) = max(Hmag(Passband)) ...
                      - min(Hmag(Passband));

    % Minimum attenuation in the stopband
```

```
    StopbandAttenuation(i) = -max(Hmag(Stopband));
end

PerformanceTable = table(WindowNames, ...
    PassbandRipple, StopbandAttenuation, ...
    'VariableNames', ...
    {'Window', 'PassbandRipple_dB', 'StopbandAttenuation_dB'});

fprintf('\n=============================================\n');
```

```
=============================================
```

```
fprintf('       FIR FILTER PERFORMANCE ANALYSIS\n');
```

```
       FIR FILTER PERFORMANCE ANALYSIS
```

```
fprintf('=============================================\n');
```

```
=============================================
```

```
disp(PerformanceTable);
```

Window	PassbandRipple_dB	StopbandAttenuation_dB
"Rectangular"	0.63743	25.37
"Hamming"	0.112	39.886
"Hann"	0.15184	39.249
"Blackman"	0.4415	26.177

```
%% 12. FILTER ORDER COMPARISON

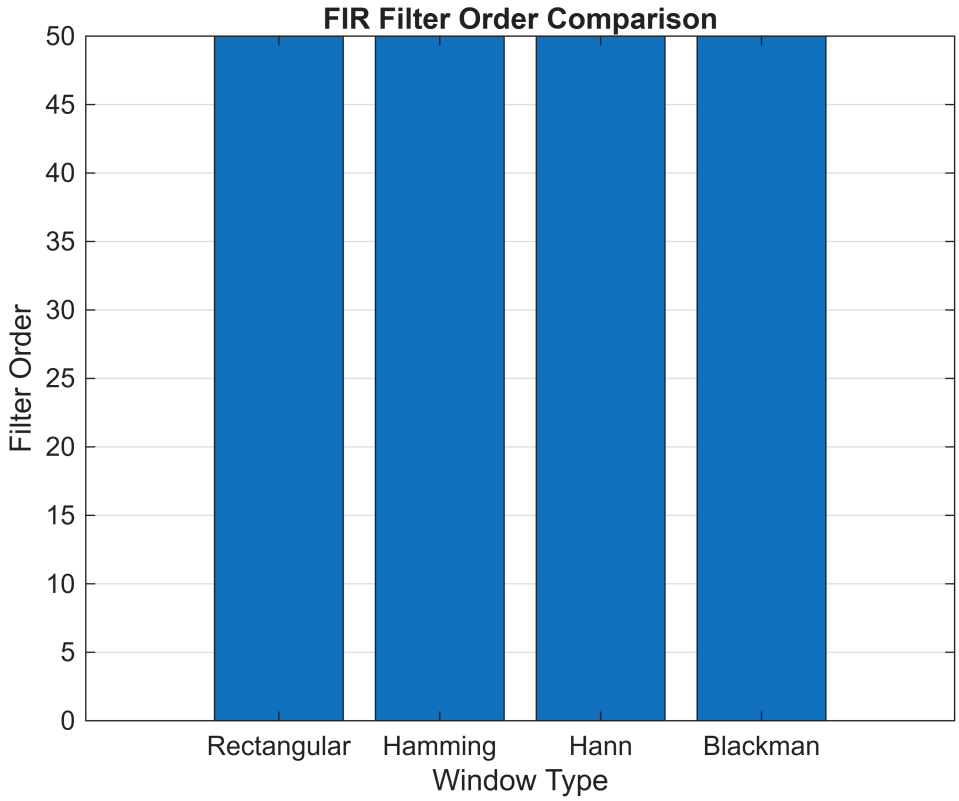
figure;

bar([N-1 N-1 N-1 N-1]);

set(gca, 'XTick', 1:4);
set(gca, 'XTickLabel', ...
    {'Rectangular', 'Hamming', 'Hann', 'Blackman'});

ylabel('Filter Order');
xlabel('Window Type');
title('FIR Filter Order Comparison');

grid on;
```




```
%% 13. FILTER A TEST SIGNAL

t = 0:1/Fs:0.1-1/Fs;

% Input signal containing passband and stopband frequencies
x = sin(2*pi*50*t) + sin(2*pi*300*t);

% Filter the signal using each FIR design
y_rect = filter(b_rect, 1, x);
y_hamming = filter(b_hamming, 1, x);
y_hann = filter(b_hann, 1, x);
y_blackman = filter(b_blackman, 1, x);

figure;

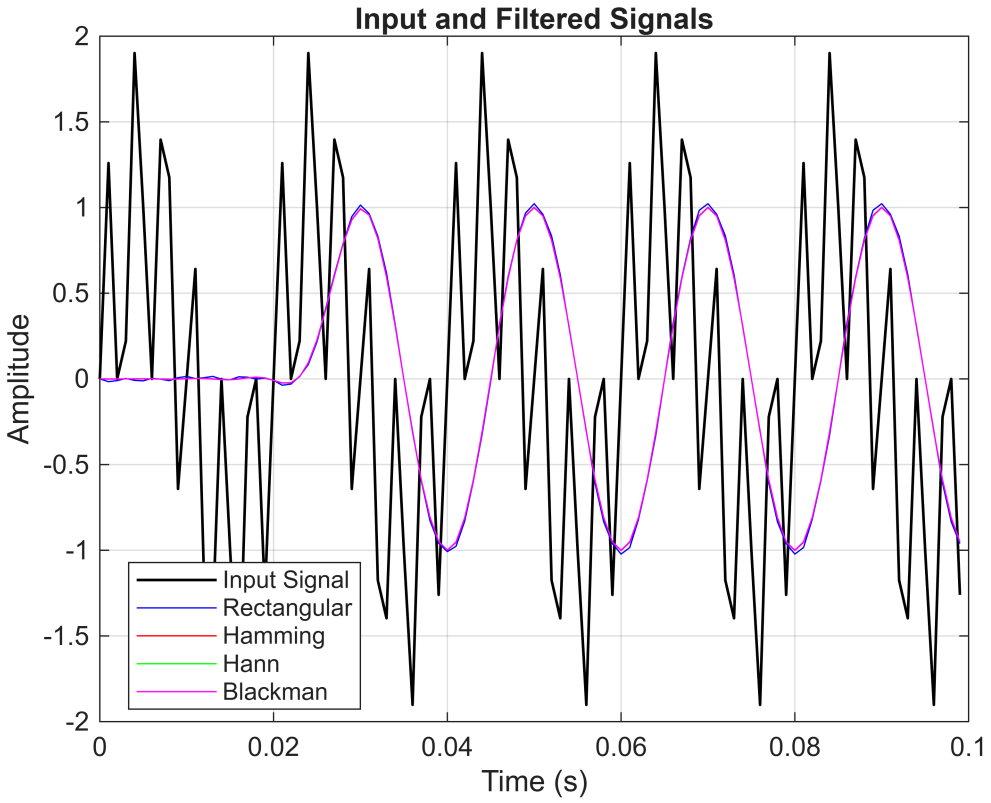
plot(t, x, 'k', 'LineWidth', 1);
hold on;
plot(t, y_rect, 'b');
plot(t, y_hamming, 'r');
plot(t, y_hann, 'g');
plot(t, y_blackman, 'm');

grid on;
xlabel('Time (s)');
ylabel('Amplitude');
```

```
title('Input and Filtered Signals');

legend('Input Signal', 'Rectangular', 'Hamming', ...
       'Hann', 'Blackman', 'Location', 'best');

hold off;
```




```
%% ============================================================
% END OF EXPERIMENT
%% ============================================================
```